Automatic Solar Tracker System

using Arduino UNO, LDR Sensors & Servo Motor

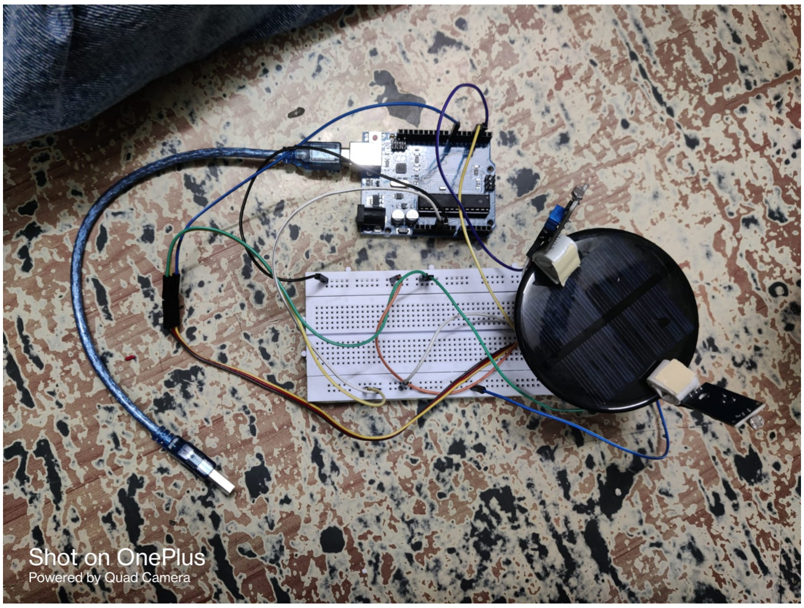

Fig 1: Assembled hardware prototype

Project Report

1. Introduction

This project is an automatic solar tracking system that continuously orients a small solar panel towards the brightest available light source. Two Light Dependent Resistors (LDRs) are mounted on either side of the panel and compare the amount of light falling on each side. An Arduino UNO reads both LDR values and rotates a servo motor, on which the solar panel is mounted, to gradually align the panel with the light source, improving its light-capturing efficiency compared to a fixed panel.

1.1 Objective

To build a simple closed-loop light-tracking mechanism using two LDR sensors and a single servo motor, where the Arduino continuously compares light intensity on the left and right sides and adjusts the panel's angle to always face the stronger light source.

2. Components Used

S.No	Component	Quantity	Purpose
1	Arduino UNO	1	Main controller
2	LDR (Light Dependent Resistor)	2	Sense light intensity (left & right)
3	10k ohm Resistor	2	Voltage divider with each LDR
4	Servo Motor (SG90 type)	1	Rotates the solar panel
5	Mini Solar Panel	1	Mounted on servo shaft, the tracked load
6	Breadboard	1	Building the LDR voltage divider circuit
7	Jumper Wires	As required	Connections between modules
8	USB Cable	1	Power/programming the Arduino

3. Circuit Diagram & Connections

Each LDR forms a voltage divider with a 10k resistor on the breadboard; the midpoint of each divider feeds an analog input on the Arduino. Brighter light on an LDR lowers its resistance and changes the analog voltage read at A0 (left) and A1 (right). The Arduino compares these two readings and drives the servo, on which the panel is mounted, to reduce the difference.

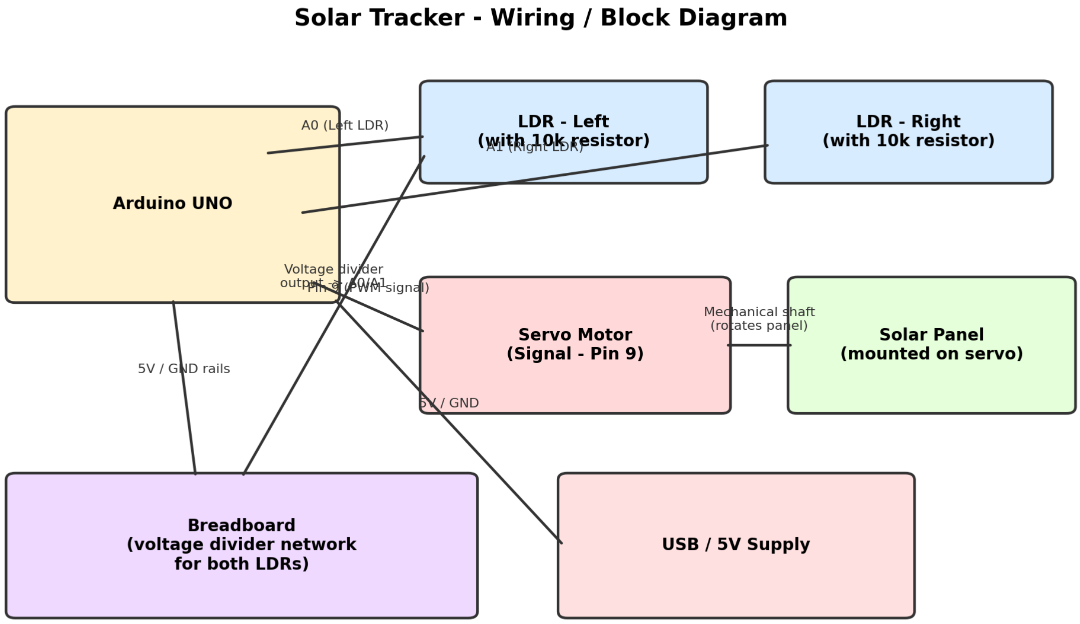

Fig 2: Block wiring diagram of the connections

3.1 Pin Connection Table

Arduino Pin	Connects To	Function
A0	Left LDR voltage divider output	Reads left-side light intensity
A1	Right LDR voltage divider output	Reads right-side light intensity
Pin 9	Servo signal wire	PWM control of servo angle (0-180 deg)
5V	Servo VCC, LDR voltage dividers	Power supply
GND	Servo GND, LDR voltage dividers	Common ground

4. Working Principle

At startup, the servo is set to a neutral 90-degree position. In the main loop, the Arduino continuously reads the analog voltage from both the left LDR (A0) and right LDR (A1) divider circuits.

- If the left LDR reading is significantly greater than the right (by more than a fixed threshold), the servo angle is decreased by one degree, rotating the panel toward the left.
- If the right LDR reading is significantly greater than the left, the servo angle is increased by one degree, rotating the panel toward the right.
- If both readings are close to each other (within the threshold), the panel position is held steady, meaning it is already facing the brightest source.
- The servo angle is constrained between 0 and 180 degrees, and a small 20 ms delay is used between readings so the panel moves smoothly rather than jittering.

This simple proportional-style comparison logic repeats continuously, so the panel keeps re-aligning itself as the light source (such as the sun) moves across the sky, or as a torch/lamp is moved near the sensors during testing.

5. Source Code (Arduino IDE)

The code below runs on the Arduino UNO. It reads both LDRs, compares their values, and adjusts the servo angle to track the brighter light source.

```
#include <Servo.h>

Servo solar;

const int leftLDR = A0;
const int rightLDR = A1;

int pos = 90;
int threshold = 50;

void setup() {
solar.attach(9);
solar.write(pos);
}

void loop() {

int leftValue = analogRead(leftLDR);
int rightValue = analogRead(rightLDR);

if (leftValue > rightValue + threshold) {
pos--;
}
else if (rightValue > leftValue + threshold) {
pos++;
}

pos = constrain(pos, 0, 180);

solar.write(pos);

delay(20);
}
```

6. Setup & Connection Steps

1. On the breadboard, build two voltage-divider circuits: each LDR in series with a 10k resistor between 5V and GND, with the midpoint tapped off to an analog pin.
2. Connect the left LDR divider's midpoint to Arduino pin A0, and the right LDR divider's midpoint to pin A1.
3. Connect the servo motor's signal wire to Arduino pin 9, and its power wires to 5V and GND.
4. Mount the solar panel on the servo horn/shaft so it rotates with the servo, and fix the two LDRs on either side of the panel facing outward.
5. Upload the Arduino sketch above using the Arduino IDE (the built-in Servo library is used, no extra installation needed).
6. Power the Arduino via USB; the panel will start at the center position and rotate left/right to follow the strongest light source.
7. Test by shining a torch or moving a lamp near either LDR; the panel should turn to face it.

7. Conclusion

This project successfully demonstrates a basic single-axis automatic solar tracking system using an Arduino UNO, two LDR sensors, and a servo motor. By continuously comparing light intensity from two directions and adjusting the panel's angle accordingly, the system keeps the solar panel more closely aligned with the light source than a fixed, non-tracking panel would be, which can meaningfully improve energy capture over the course of a day.

The design can be further improved by adding a second axis (dual-servo, dual-LDR-pair) for full elevation and azimuth tracking, using more LDRs for finer directional resolution, adding a data logger or display to monitor panel output voltage/current, or replacing the fixed threshold logic with a smoother PID-based control loop. Overall, the project provides a good practical foundation for learning sensor-based feedback control and renewable-energy-related embedded systems.